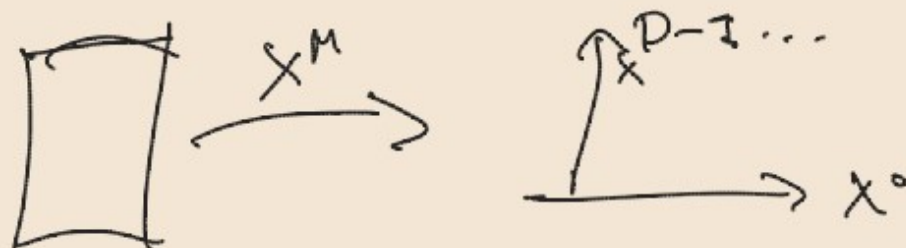


The Dynamics:

The World sheet has coordinates

$$\xi^\alpha \equiv (\tau, \sigma), \quad \xi^0 = \tau, \quad \xi^1 = \sigma$$



The Worldsheet is Embedded into M^D via $X^\mu(\xi^\alpha)$.

$$X: \Sigma \rightarrow M^D$$

(i) Open string: $0 \leq \sigma \leq \pi$

(ii) Closed string: $-\pi \leq \sigma \leq \pi$ w/ B.C.'s
 $X^\mu(\tau, -\pi) = X^\mu(\tau, \pi)$

• Similar to the ^{Point} particle which has
 $S_{pp} = -m \int d\tau = -m \int dt \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$

We have a generalisation of this Action for p-branes:

$$S = -T_p \int d\mu^p, \quad T_p: \text{p-brane tension}$$

$$d\mu^p \equiv \sqrt{-\det(G_{\alpha\beta})} d^{p+1}\sigma$$

Where $G_{\alpha\beta}$ is the induced metric

$$G_{\alpha\beta}^I = g_{\mu\nu}(X) \partial_\alpha X^\mu \partial_\beta X^\nu$$

$$\Rightarrow \partial_\alpha X^\mu \equiv \partial x^\mu / \partial \xi^\alpha$$

* Here, $M^D \Rightarrow g_{\mu\nu}(X) = \eta_{\mu\nu}$ thus

$$S_{NG} = -\frac{1}{2\alpha\alpha'} \int d^2 \xi \sqrt{-\det(\eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu)}$$

- Note: This determinant is over the α, β coordinates and so we would expect

$$\eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu = \eta_{\mu\nu} \begin{pmatrix} \partial_0 X^\mu \partial_0 X^\nu & \partial_0 X^\mu \partial_1 X^\nu \\ \partial_1 X^\mu \partial_0 X^\nu & \partial_1 X^\mu \partial_1 X^\nu \end{pmatrix}$$

$$\Rightarrow \det(G_{\alpha\beta}) = \partial_0 X^\mu \partial_0 X_\mu \partial_1 X^\nu \partial_1 X_\nu - \partial_0 X^\mu \partial_1 X_\mu \partial_1 X^\nu \partial_0 X_\nu$$

- If we move from $\xi^a \mapsto \xi^a + d\xi^a$ on Σ we have a change in M^D given by $X^\mu(\xi^a) \mapsto X^\mu(\xi^a) + d\xi^a \partial_a X^\mu$

The distance in M^D is given by $\eta_{\mu\nu} (d\xi^a \partial_a X^\mu) (d\xi^b \partial_b X^\nu)$

However Note this is simply the induced metric

$$G_{\alpha\beta} = \eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu$$

$$\text{Since } d\xi^a d\xi^b G_{\alpha\beta} = g_\Sigma$$

* This is very childish, more precisely $X: \Sigma \rightarrow M^D$

$$X^*: T^* M^D \rightarrow T^* \Sigma$$

$$(X^* \eta_{\mu\nu})(X, Y) = \eta_{\mu\nu}(X_* X, X_* Y)$$

$$(X_* X)(t) = \frac{d}{dt} X(t) = \dot{X}^\mu \frac{\partial}{\partial X^\mu} (t, X) = \dot{X}^\mu \frac{\partial \xi^a}{\partial X^\mu} \partial_a t$$

$\therefore$

$$(X^* \eta)(X, Y) = \dot{X}^\alpha (X^* \eta)_{\alpha\beta} \dot{Y}^\beta$$

$$L(\dot{X}) = \eta(X, X) = \eta_{\mu\nu} \dot{X}^\mu \dot{X}^\nu$$

Where $(X^* \Sigma)(f) = \Sigma(f \circ X)$

$\Sigma^\alpha \partial_\alpha (f \circ X)$

$f: M^D \rightarrow \mathbb{R} \quad X: \Sigma \rightarrow M^D$

$f \circ X: \Sigma \rightarrow \mathbb{R}$

$\Sigma^\alpha \frac{\partial X^\mu}{\partial \xi^\alpha} \partial_\mu f = (X^* \Sigma)(f)$

$\mathcal{N}(\Sigma^\alpha \frac{\partial X^\mu}{\partial \xi^\alpha} \partial_\mu f, \Sigma^\beta \frac{\partial X^\nu}{\partial \xi^\beta} \partial_\nu f)$
 $= \Sigma^\alpha \left[\frac{\partial X^\mu}{\partial \xi^\alpha} \eta_{\mu\nu} \frac{\partial X^\nu}{\partial \xi^\beta} \right] \Sigma^\beta$

$\therefore$ We conclude

$(X^* \mathcal{N})_{\alpha\beta} = G_{\alpha\beta}^I = \frac{\partial X^\mu}{\partial \xi^\alpha} \eta_{\mu\nu} \frac{\partial X^\nu}{\partial \xi^\beta}$

• Introduce the 2D vielbein e_a^α

$e_a^\alpha \eta_{ab} e_b^\beta = G_{\alpha\beta}^I$

a : Lorentz indices on Σ

α : Σ world sheet indices

$\eta_{ab} = \text{diag}(-1, 1)$

• On the world sheet we have tangent vectors

$e_a^\alpha \partial_\alpha^\xi$

for which the area is given by

$\det(e_a^\alpha \partial_\alpha^\xi \partial_\alpha^\eta)$

* Note for a static string, the kinetic term vanishes (i.e. all ∂_τ terms)

and $P=0$ so the Hamiltonian

$H = P\dot{q} - L \Rightarrow H = -L$

$\therefore S = T \int d\tau d\sigma H \rightarrow T \int d\sigma H$

since S is unitless, we find

Symmetries:

(Diffeomorphism)

(i) Reparametrizations: $\xi^\alpha \mapsto \xi'^\alpha(\xi)$

$$x^\mu(\xi') = x^\mu(\xi)$$

Infinitesimally: $\delta \xi^\alpha = \xi'^\alpha - \xi^\alpha$

$$\text{i.e. } \xi'^\alpha = \xi^\alpha + \delta \xi^\alpha$$

$$\therefore x^\mu(\xi^\alpha + \delta \xi^\alpha) = x^\mu(\xi) + \partial_\alpha x^\mu \delta \xi^\alpha$$

$$\Rightarrow \delta x^\mu = x'^\mu(\xi) - x^\mu(\xi) = -\delta \xi^\alpha \partial_\alpha x^\mu$$

In particular we note that the induced metric is also affected

$$G_{\alpha\beta}(\xi) = (\partial_\alpha \xi'^\gamma / \partial \xi^\alpha) (\partial_\beta \xi'^\delta / \partial \xi^\beta) G_{\gamma\delta}(\xi')$$

(a) $\xi^\alpha \mapsto \xi'^\alpha(\xi)$, $\delta \xi^\alpha = \xi'^\alpha - \xi^\alpha$

(b) $x^\mu(\xi) = x'^\mu(\xi')$
 $\delta x^\mu = -\delta \xi^\alpha \partial_\alpha x^\mu$

(c)

* Rep. invariance of the Nambu-Goto Action:

$$S = T \int d^2 \xi \sqrt{-\det(\partial_\alpha x^\mu \partial_\beta x^\nu \eta_{\mu\nu})}$$

Note $S \rightarrow S'$

$$\frac{\partial x^\mu}{\partial \xi'^\alpha} \frac{\partial x^\nu}{\partial \xi'^\beta} = \frac{\partial \xi^\gamma}{\partial \xi'^\alpha} \frac{\partial \xi^\delta}{\partial \xi'^\beta} \partial_\gamma x^\mu \partial_\delta x^\nu \eta_{\mu\nu}$$

Since $x^\mu(\xi') = x^\mu(\xi)$

Putting this together we find

$$S' = T \int d^2 \xi' \sqrt{-\det \left[\frac{\partial \xi^\gamma}{\partial \xi'^\alpha} \right]^2 \det [\partial_\gamma x^\mu \partial_\delta x^\nu \eta_{\mu\nu}]}$$

$$\det[\partial \xi^{\mu} / \partial \xi'^{\nu}] \equiv J$$

The Measure transforms as

$$d^2 \xi' = |\partial \xi' / \partial \xi| d^2 \xi = J^{-1} d^2 \xi$$

$$\therefore S' = T \int d^2 \xi J^{-1} J \sqrt{-\det(\partial_{\alpha} x^{\mu} \partial_{\beta} x_{\mu})}$$

(ii) Rigid Poincaré Invariance:

$$x^{\mu'}(\xi) = \Lambda^{\mu}_{\nu} x^{\nu}(\xi) + a^{\mu}$$

$$\Lambda^{\mu}_{\rho} \eta_{\mu\nu} \Lambda^{\nu}_{\tau} = \eta_{\rho\tau}$$

$$\text{Note } \eta_{\mu\nu} = -\eta_{\nu\mu}$$

$$\delta_{\rho} h_{\alpha\beta} = 0$$

Infinitesimally

$$\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu} + a^{\mu}_{\nu}, \quad \text{as such}$$

$$\begin{aligned} \delta_{\rho} x^{\mu} &\equiv x^{\mu'}(\xi) - x^{\mu} = (\delta^{\mu}_{\nu} + a^{\mu}_{\nu}) x^{\nu} + a^{\mu} - x^{\mu} \\ &= a^{\mu}_{\nu} x^{\nu} + a^{\mu} \end{aligned}$$

$$|a^{\mu}_{\nu}| \ll 1$$

We find $\eta_{\mu\nu} = -\eta_{\nu\mu}$ by requiring that $\delta(\eta_{\mu\nu} x^{\mu} x^{\nu}) = 0$

* Since Quantizing $\sqrt{\quad}$ is particularly difficult, we introduce a "verlben" equivalent of e for the point particle: a metric $g_{\alpha\beta}$

Polyakov Action:

$$S = -\frac{1}{4\pi\alpha'} \int d^2 \xi \sqrt{-g} g^{\alpha\beta} \partial_{\alpha} x^{\mu} \partial_{\beta} x^{\nu} \eta_{\mu\nu}$$

$g_{\alpha\beta}$: 2D Metric NATURALLY on Σ_1 .
 when is Auxiliary.

- Since $\mathcal{L}_{\text{NLS}}$ is from M^D this Action is D-scalar fields x^M all coupled to two-dimensional gravity $g_{\alpha\beta}$ on Σ_1 .

* This action, unsurprisingly, contains more sym than N.G. Action

(i) Diffeomorphisms: $x^M(\xi') = x^M(\xi)$

$$g'_{\alpha\beta}(\xi') = \left(\frac{\partial \xi^\gamma}{\partial \xi'^\alpha} \right) \left(\frac{\partial \xi^\delta}{\partial \xi'^\beta} \right) g_{\gamma\delta}(\xi)$$

(ii) Poincare: (As above)

(iii) Weyl:

$$g'_{\alpha\beta} = e^{\phi(\sigma, \epsilon)} g_{\alpha\beta}, \quad x^M(\xi') = x^M(\xi)$$

E.O.M.: $S = \tau \int d^2 \xi \sqrt{-g} g^{\alpha\beta} \partial_\alpha x^M \partial_\beta x^N \mathcal{L}_{\text{NLS}}$

Our Lagrangian (density)

$$\mathcal{L} = \tau \sqrt{-g} g^{\alpha\beta} \partial_\alpha x^M \partial_\beta x^N \mathcal{L}_{\text{NLS}}$$

$$\therefore 0 = \delta S = \int_\sigma^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} \delta g_{\alpha\beta} + \frac{\partial \mathcal{L}}{\partial \partial_\alpha x^M} \delta(\partial_\alpha x^M) \right]$$

The second term expands over the α index:

$$\frac{\partial \mathcal{L}}{\partial \partial_\alpha x^M} \delta(\partial_\alpha x^M) = \frac{\partial \mathcal{L}}{\partial \partial_\tau x^M} \delta(\partial_\tau x^M) + \frac{\partial \mathcal{L}}{\partial \partial_\sigma x^M} \delta(\partial_\sigma x^M)$$

→ Require that $[\delta x^1 = \delta x^2 = \delta g_{\alpha\beta} = 0]$

$$0 = \int_\sigma^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left\{ \frac{\partial \mathcal{L}}{\partial g} \delta g_{\alpha\beta} + \frac{\partial \mathcal{L}}{\partial \partial_\tau x^M} \delta(\partial_\tau x^M) + \frac{\partial \mathcal{L}}{\partial \partial_\sigma x^M} \delta(\partial_\sigma x^M) \right\}$$

Integrate by Parts so that

$$\textcircled{a} \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \partial_\tau (\delta X^\mu) = \iint d\sigma d\tau \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \delta X^\mu \right) \\ = \int_a^\pi d\sigma \left[\frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \delta X^\mu \right]_{\tau_1}^{\tau_2} - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \partial_\tau \frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \delta X^\mu$$

$$\text{But } \delta X^\mu(\tau_1) = \delta X^\mu(\tau_2) = 0$$

$$\therefore \textcircled{a} = - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \partial_\tau \frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \delta X^\mu$$

$$\textcircled{b} = \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial (\partial_\sigma X^\mu)} \delta X^\mu \right]_{\sigma=a}^{\sigma=\pi} - \int_a^\pi d\sigma \int_{\tau_1}^{\tau_2} d\tau \left(\partial_\sigma \frac{\partial \mathcal{L}}{\partial (\partial_\sigma X^\mu)} \delta X^\mu \right)$$

$$\therefore 0 = \iint_{\tau_1}^{\tau_2} d\sigma d\tau \left\{ \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} \delta g_{\alpha\beta} - \partial_\sigma \left(\frac{\partial \mathcal{L}}{\partial (\partial_\sigma X^\mu)} \delta X^\mu \right) - \partial_\tau \left(\frac{\partial \mathcal{L}}{\partial (\partial_\tau X^\mu)} \delta X^\mu \right) \right\} \\ + \int_{\tau_1}^{\tau_2} d\tau \left[\frac{\partial \mathcal{L}}{\partial (\partial_\sigma X^\mu)} \delta X^\mu \right]_{\sigma=a}^{\sigma=\pi}$$

Case (i): $\delta X^\mu = 0$, $\delta g_{\alpha\beta}$: arbitrary

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} = 0$$

Case (ii): $\delta g_{\alpha\beta} = 0$, δX^μ : arbitrary up to $\delta X^\mu(\pi) = \delta X^\mu(a) = 0$

$$\textcircled{b} \quad \partial_\alpha \pi^{\alpha\mu} = 0 \quad \text{where} \quad \pi^{\alpha\mu} = \frac{\partial \mathcal{L}}{\partial (\partial_\alpha X^\mu)}$$

$$\textcircled{a} \quad \frac{\partial \mathcal{L}}{\partial g_{\alpha\beta}} = 0 = T \sqrt{-g} \partial^\alpha X^\mu \partial^\beta X_\mu + \dots$$

$$\text{note } \delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g^{\alpha\beta} \delta g_{\alpha\beta} = -\frac{1}{2} \sqrt{-g} g^{\alpha\beta} \partial_\gamma \delta g_{\alpha\beta}$$

$$\partial \mathcal{L} / \partial g_{\alpha\beta} = \sqrt{-g} \partial^\alpha X^\mu \partial^\beta X_\mu - \frac{1}{2} g^{\alpha\beta} \partial_\gamma X^\mu \partial_\delta X_\mu g^{\gamma\delta}$$

$$\therefore 0 = \partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} g_{\alpha\beta} g^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X_\mu$$

$$\textcircled{b} \partial_\tau \left(\frac{\partial \mathcal{L}}{\partial \partial_\tau x^\mu} \right) + \partial_\sigma \left(\frac{\partial \mathcal{L}}{\partial \partial_\sigma x^\mu} \right)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \partial_\sigma x^\mu} &= \frac{\partial}{\partial \partial_\sigma x^\mu} \left[-T \sqrt{-g} g^{\alpha\beta} \eta^{\mu\nu} \partial_\alpha x_\mu \partial_\beta x_\nu \right] \\ &= 2 \sqrt{-g} T g^{\alpha\beta} \eta^{\mu\nu} \delta_\alpha^\sigma \delta_\mu^\lambda \\ &= (2 \sqrt{-g} T) (g^{\sigma\beta} \eta^{\lambda\nu} \partial_\beta x_\nu) \end{aligned}$$

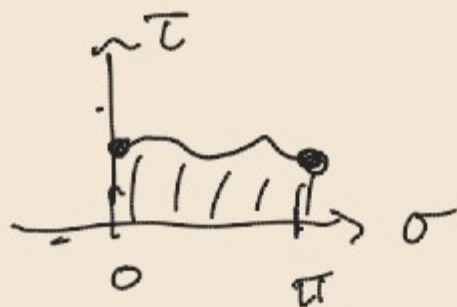
$$\therefore \partial_\alpha \pi^{\alpha\mu} = \partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta x^\mu) = 0$$

* For the closed string: $x^\mu(\tau, \sigma) = x^\mu(\tau, \sigma + 2\pi)$
 so $\delta x^\mu(\tau, 0) = \delta x^\mu(\tau, 2\pi)$ and so the Boundary term vanishes

* For the open string, we take

$$\partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta x^\mu) = 0$$

to hold at the Boundary $\sigma=0, \sigma=\pi$



→ There is no σ restriction on δx^μ this implies $\delta S = 0 \iff$

$$\pi^{\lambda\mu} \equiv \frac{\partial \mathcal{L}}{\partial \partial_\lambda x^\mu} = 0 \quad \text{at } \sigma=0 \text{ and } \sigma=\pi$$

We have, from above

$$\pi^{\lambda\mu} = \sqrt{-g} g^{\lambda\beta} \partial_\beta x^\mu = 0$$

$$\therefore \sqrt{-g} g^{\lambda\beta} \partial_\beta x^\mu = 0 \quad \text{at } \sigma=0, \sigma=\pi$$

$$\sqrt{-g} g^{2B} \partial_B X^m \Big|_{\sigma=2\pi, \tau} = 0$$

This is really two separate equations:

→ There is no relationship between the two ends of the string.

* Now, from the gauged E.O.M we know

$$\frac{1}{2} g_{\alpha\beta} f^{-1} = \partial_\alpha X^m \partial_\beta X_m \quad \text{or}$$

$$g_{\alpha\beta} = 2f \partial_\alpha X^m \partial_\beta X_m, \quad f^{-2} \equiv g^{\alpha\beta} \partial_\alpha X^m \partial_\beta X_m$$

$$\text{Thus, } g = \det(g_{\alpha\beta}) = (2f)^2 \det(\partial_\alpha X^m \partial_\beta X^m \eta_{mn}) \\ = 4f^2 \det(G_{\alpha\beta})$$

→ Side Note: $g_{\alpha\beta}$ is Conformally Eq. to the induced metric $G_{\alpha\beta}$

$$\sqrt{-g} = 2f \sqrt{-\det(G_{\alpha\beta})}$$

$$\text{or } \sqrt{-g} f^{-1} = 2 \sqrt{-\det(G_{\alpha\beta})}$$

$$\sqrt{-g} g^{\alpha\beta} \partial_\alpha X^m \partial_\beta X^m = 2 \sqrt{-\det(G_{\alpha\beta})}$$

↳ This equivalence allows us to understand the Equivalence of N.G.'s, Pol. actions at the Classical level.

Gauge Choices:

Riemann surface: 2D surface w/ a Euclidean metric

↳ On any Riemann surface one can always locally find a $f^{\mu} \phi(\xi)$ s.t.

$$g_{\alpha\beta}(\xi) = \eta_{\alpha\beta} e^{\phi(\xi)}$$

→ Physics: Counting D.O.F. $g_{\alpha\beta}$ is

symmetric and $g^{\alpha\beta}g_{\alpha\beta} = 2$

$$\therefore g_{\alpha\beta} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \Rightarrow \alpha + \delta = 2$$

So $g_{\alpha\beta}$ has 3 D.O.F. However

$g_{\alpha\beta}^c = e^{\phi(\xi)} \eta_{\alpha\beta}$ has 1 D.O.F. due to $\phi(\xi)$

* Note: Cannot make a Gauge choice Before we have taken the variation of S_r

→ Impose gauge choices on the E.O.M

$$\Leftrightarrow \partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta x^\mu) = 0$$

In the Conformal Gauge

$$g_{\alpha\beta}^c = e^{\phi(\xi)} \eta_{\alpha\beta}$$

$$\partial_\alpha (e^{\phi} e^{-\phi} \eta^{\alpha\beta} \partial_\beta x^\mu) = \partial_\alpha (\eta^{\alpha\beta} \partial_\beta x^\mu) = 0$$

traceless

• This leads to the wave-equation

$$\partial^2 x^\mu = 0$$

• The Second E.O.M: $\partial_\alpha x^\mu \partial_\beta x_\mu \cdot \frac{1}{2} g_{\alpha\beta} g^{\delta\lambda} \partial_\delta x^\mu \partial_\lambda x_\mu$ becomes

$$\partial_\alpha x^\mu \partial_\beta x_\mu - \frac{1}{2} e^{\phi} \eta_{\alpha\beta} e^{-\phi} \eta^{\delta\lambda} \partial_\delta x^\mu \partial_\lambda x_\mu = 0$$

$$T_{\alpha\beta} \equiv \partial_\alpha x^\mu \partial_\beta x_\mu - \frac{1}{2} \eta_{\alpha\beta} \partial_\alpha x^\mu \partial_\alpha x_\mu = 0$$

* We note that $T_{\alpha\beta}$ is also traceless:

$$\eta^{\alpha\beta} T_{\alpha\beta} = 0 \quad (\text{trivially})$$

$$T_{00} = T_{11} = \frac{1}{2} (\dot{x}^\mu \dot{x}^\mu + x^{\mu'} x^{\nu'} \eta_{\mu\nu'})$$

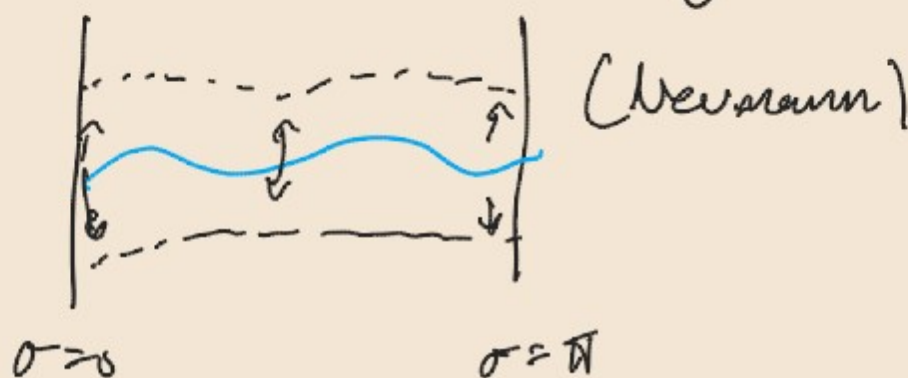
$$T_{01} = T_{10} = \dot{x}^\mu x^{\nu'} \eta_{\mu\nu'}$$

$$D_{\alpha\beta} = D_{\beta\alpha} = \frac{1}{2} (\dot{x}^\mu \dot{x}^\mu - x^{\mu'} x^{\nu'} \eta_{\mu\nu'})$$

(i) Open w/ Neumann: Set derivative of X^μ by σ to vanish

$$\partial_\sigma X^\mu(\tau, \sigma + \pi) = \partial_\sigma X^\mu(\tau, \sigma) = 0$$

→ No momentum is flowing through the ends of the string



• Neumann B.C's preserve Poincaré invariance

$$\begin{aligned} \text{Proof } \partial_\sigma(X^\mu)|_{\sigma=0,\pi} &= \partial_\sigma(a_\nu^\mu X^\nu + a^\mu)|_{\sigma=0,\pi} \\ &= a_\nu^\mu (\partial_\sigma X^\nu)|_{\sigma=0,\pi} = 0 \quad \checkmark \end{aligned}$$

(ii) Open w/ Dirichlet:

$$X^\mu(\tau, \sigma=0) \equiv X_0^\mu \in \mathbb{R} \text{ (constant)}$$

$$X^\mu(\tau, \sigma=\pi) \equiv X_\pi^\mu \in \mathbb{R} \text{ (constant)}$$

• Dirichlet B.C's Do NOT Preserve Poincaré invariance

$$\begin{aligned} \text{Proof: } X^\mu|_{\sigma=0,\pi} &= (a_\nu^\mu X^\nu + a^\mu)|_{\sigma=0,\pi} \\ &= a_\nu^\mu X_{0,\pi}^\nu + a^\mu \\ &\neq X_{0,\pi}^\mu \end{aligned}$$

* The conjugate momentum $P^\mu = \frac{2\pi}{\alpha'} \partial_\tau X^\mu$

$$P^\mu = \dot{X}^\mu / 2\pi\alpha'$$

$$\begin{aligned} \{X^\mu(\sigma), X^\nu(\sigma')\}_{P.B} &= 0 = \{P^\mu(\sigma), P^\nu(\sigma')\}_{P.B} \\ \{X^\mu(\sigma), P^\nu(\sigma')\}_{P.B} &= \eta^{\mu\nu} \delta(\sigma - \sigma') \end{aligned}$$

(i) Closed String: (τ, σ) , $-\pi \leq \sigma \leq \pi$

E.O.M $\partial^2 x^\mu = 0$ subject to B.C.'s

$$x^\mu(\tau, \sigma) = x^\mu(\tau, \sigma + 2\pi)$$

$$(-\partial_\tau^2 + \partial_\sigma^2) x^\mu(\tau, \sigma) = 0$$

We decompose $x^\mu(\tau, \sigma) = A(\sigma) T^\mu(\tau)$

Q: How do I know where μ goes? in A or T ? $\rightarrow$ I don't think it matters...

$$(-\partial_\tau^2 + \partial_\sigma^2)(A(\sigma) T^\mu(\tau))$$

$$(\partial_\sigma^2 A) T^\mu(\tau) - A(\sigma) \partial_\tau^2 T^\mu(\tau) = 0$$

$$\partial_\sigma^2 A(\sigma) = -\lambda^2 A(\sigma)$$

$$A(\sigma) = \tilde{A} e^{i\lambda\sigma} + \tilde{B} e^{-i\lambda\sigma}$$

$$A(\sigma) = \tilde{A} + \tilde{B} = \tilde{A} e^{i\lambda 2\pi} + \tilde{B} e^{-i\lambda\sigma}$$

$$\cos(\lambda 2\pi) = 1$$

$$\lambda 2\pi = n 2\pi$$

$$\therefore \lambda = n \in \mathbb{Z}$$

$$A_n(\sigma) = \tilde{A}_n e^{in\sigma}, \quad \forall n \in \mathbb{Z}$$

If we then sum over all possible states

$$x^\mu(\sigma, \tau) = \sum_{n=-\infty}^{\infty} e^{in\sigma} x_n^\mu(\tau)$$

$\rightarrow$ To ensure $x^\mu(\sigma, \tau)$ is real we have

$$(x_n^\mu(\tau))^* = (x_{-n}^\mu(\tau))$$

$\rightarrow$ Plugging this into $(-\partial_\tau^2 + \partial_\sigma^2) x^\mu(\sigma, \tau) = 0$

$$(-\partial_\tau^2 + \partial_\sigma^2) \sum_{n=-\infty}^{\infty} e^{in\sigma} \dot{X}_n^\mu(\tau) = 0$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} \left(-\ddot{X}_n^\mu(\tau) e^{in\sigma} + (in)^2 e^{in\sigma} X_n^\mu(\tau) \right) \\ = \sum_{n=-\infty}^{\infty} e^{in\sigma} \left[\ddot{X}_n^\mu(\tau) + n^2 X_n^\mu(\tau) \right] = 0$$

$$\text{or } \boxed{\ddot{X}_n^\mu(\tau) = -n^2 X_n^\mu(\tau)}$$

(As expected...)

• The lowest mode $n=0$:

$$\ddot{X}_0^\mu(\tau) = 0$$

Is the Center of Mass coordinate

$$X_0^\mu(\tau) = \tilde{x}^\mu + p^\mu \tau$$

* To get to the Point of Quantization we introduce the Creation & ann. operators (Much like the Harmonic oscillator):

$$\gamma_{-}^\mu(\tau, \sigma) \equiv \frac{1}{\sqrt{2\alpha'}} \left(\tilde{X}^\mu(\tau, \sigma) + X^\mu(\tau, \sigma) \right)$$

$$\gamma_{+}^\mu(\tau, \sigma) \equiv \frac{1}{\sqrt{2\alpha'}} \left(\tilde{X}^\mu(\tau, \sigma) - X^\mu(\tau, \sigma) \right)$$

(Compare to $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$, $a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$)

* To solve for $\gamma_{\pm}^\mu(\tau, \sigma)$, note

$$\gamma_{\pm}^\mu(0, \sigma) = \frac{1}{\sqrt{2\alpha'}} (\partial_\tau \pm \partial_\sigma) X^\mu(\tau, \sigma)$$

Thus, we have the first order

$$(\partial_\sigma - \partial_\tau) \gamma_{\pm}^\mu(\tau, \sigma) = 0$$

$$\gamma_{\pm}^\mu(\tau, \sigma) = \alpha^\mu(\tau) A(\sigma)$$

Subject to $A(0) = A(2\pi)$

$$A'(\sigma) \alpha^\mu(\tau) = \dot{\alpha}^\mu(\tau) A(\sigma)$$

$$A(\sigma) = \tilde{A} e^{i\lambda\sigma}$$

$$\lambda = \alpha \in \mathbb{Z}$$

as before, thus

$$\dot{\alpha}_n^M(\tau) = -n \alpha_n^M(\tau)$$

Summing over all possible states

$$\mathcal{P}^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{in\sigma}$$

where all constants are absorbed in α_n^M

$$\mathcal{P}^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{-in\sigma}$$

$$\bar{\mathcal{P}}^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \bar{\alpha}_n^M(\tau) e^{in\sigma}$$

* With the conjugate momentum, we can see that $\mathcal{P}^M \equiv \dot{X}^M / 2\pi\alpha'$

[Q: It is evident that $T_{0M} \neq \mathcal{P}^M$, why is this?

• The total momentum is given by

$$p^M = \int_{-\pi}^{\pi} d\sigma \mathcal{P}^M = \int_{-\pi}^{\pi} d\sigma \dot{X}^M / 2\pi\alpha'$$

$$= \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} e^{in\sigma} \dot{X}_n^M(\tau) / 2\pi\alpha'$$

• Assuming absolute convergence... $\sum_n \int \leftrightarrow \int \sum_n$

$$\sum_{n=-\infty}^{\infty} \left(\int_{-\pi}^{\pi} d\sigma e^{in\sigma} \right) \dot{X}_n^M(\tau) = \sum_{n \neq 0} \left(\frac{2}{in} e^{in\sigma} \right) \dot{X}_n^M(\tau) \Big|_{-\pi}^{\pi}$$

$$+ 2\pi \dot{X}_0^M(\tau) / (\alpha' 2\pi)$$

$$\begin{cases} e^{in\pi} = \cos(n\pi) + i \sin(n\pi) & n \geq 0 \end{cases}$$

$$n=0 \Rightarrow 1$$

$$n \geq 1 \Rightarrow -1 = (-1)^n$$

$$n \geq 2$$

$$e^{-in\pi} = -1 \quad n \geq 0$$

$$n=1 \Rightarrow -1$$

Leaving $\boxed{P^M = \frac{\dot{X}_0^M(\tau)}{\alpha}}$ (total momentum)

* Check:

$$P^M = \frac{\dot{X}_0^M}{\alpha} = \frac{1}{\sqrt{2\alpha'}} \int_{-\pi}^{\pi} d\sigma P^M = \frac{1}{\sqrt{2\alpha'}} \int_{-\pi}^{\pi} d\sigma \mathcal{P}^M = \sqrt{\frac{2}{\alpha'}} \alpha_0^M = \sqrt{\frac{2}{\alpha'}} \bar{\alpha}_0^M$$

where $\alpha_0^M = \bar{\alpha}_0^M$

Proof:

$$\frac{1}{\sqrt{2\alpha'}} \int_{-\pi}^{\pi} d\sigma \left(\frac{1}{2\alpha'} (\dot{X}^M(\tau, \sigma) + X'^M(\tau, \sigma)) \right)$$

Using $X^M(\tau, \sigma) = \sum_{n=-\infty}^{\infty} e^{in\sigma} X_n^M(\tau)$

$$\dot{X}^M = \sum e^{in\sigma} \dot{X}_n^M(\tau), \quad X'^M = \sum i n e^{in\sigma} X_n^M(\tau)$$

$$\frac{1}{2\alpha'} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} (e^{in\sigma}) (\dot{X}_n^M(\tau) + i n X_n^M(\tau))$$

$= 0 \quad \forall n \neq 0$

$$n=0: \frac{1}{2\alpha'} \int_{-\pi}^{\pi} d\sigma \dot{X}_0^M = \frac{\dot{X}_0^M}{\alpha} \quad \checkmark$$

In terms of raising & lowering ops

$$P^M = \frac{1}{\sqrt{2\alpha'}} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{in\sigma} = \frac{2\pi}{\sqrt{2\alpha'}} \alpha_0^M(\tau)$$

$$P^M = \dot{X}_0^M / \alpha' = \sqrt{\frac{2}{\alpha'}} \alpha_0^M(\tau)$$

* Since $\mathcal{P}^M = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^M + X'^M)$ and $P^M = \frac{\dot{X}_0^M}{2\pi\alpha'}$

$$\mathcal{P}^M = \frac{1}{\sqrt{2\alpha'}} (2\pi\alpha' P^M + X'^M)$$

$$= \sqrt{2\alpha'} \pi P^M + \frac{1}{\sqrt{2\alpha'}} X'^M$$

$$\mathcal{P}^M = \sqrt{2\alpha'} \pi P^M - \frac{\alpha'^M}{\sqrt{2\alpha'}}$$

with

$$\{X^\mu, X^\nu\} = 0 = \{P^\mu, P^\nu\}$$

$$\{X^\mu, P^\nu\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

$$\Rightarrow \{\mathcal{P}^M(\sigma), \mathcal{P}^M(\sigma')\} = \left\{ \sqrt{2\alpha'} \pi P^M + \frac{1}{\sqrt{2\alpha'}} X'^M, \sqrt{2\alpha'} \pi P^M - \frac{1}{\sqrt{2\alpha'}} X'^M \right\}$$

$$= \{P^M, P^M\} - \{P, X\} + \{X, P^M\} = \{X, X\}$$

$\Rightarrow$

$$\therefore \{P^\mu(\sigma), P^\nu(\sigma')\}_{PB} = 0$$

$$\begin{aligned} \bullet \{P^\mu(\sigma), P^\nu(\sigma')\} &= \left\{ \sqrt{2\alpha'} \pi P^\mu + \frac{1}{\sqrt{2\alpha'}} X'^\mu, \sqrt{2\alpha'} P^\nu \right\} \\ &= \{P, P\} + \{X', X'\} + 2 \{X'^\mu, P^\nu\} \\ &= 2 \left\{ \frac{1}{\sqrt{2\alpha'}} \frac{d}{d\sigma} X^\mu, \sqrt{2\alpha'} \pi P^\nu \right\} \\ &= 2\pi \frac{d}{d\sigma} (\eta^{\mu\nu} \delta(\sigma - \sigma')) \end{aligned}$$

$$\therefore \{P^\mu(\sigma), P^\nu(\sigma')\} = 2\pi \eta^{\mu\nu} \frac{d}{d\sigma} (\delta(\sigma - \sigma'))$$

* Similarly we find that

$$\{\bar{P}^\mu(\sigma), \bar{P}^\nu(\sigma')\} = -2\pi \eta^{\mu\nu} \frac{d}{d\sigma} (\delta(\sigma - \sigma'))$$

• We can now find commutation relations for all of the modes α_n^μ :

$$P^\mu(\sigma, \tau) = \sum_{n=-\infty}^{\infty} \alpha_n^\mu(\tau) e^{in\sigma}$$

$$\{P^\mu, P^\nu\} = \sum_{n,m} e^{-in\sigma} e^{-im\sigma'} \{\alpha_n^\mu(\tau), \alpha_m^\nu(\tau)\}$$

Now, Recall that the delta δ has

the expansion

$$\begin{aligned} \delta(\sigma - \sigma') &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} e^{in(\sigma - \sigma')} \\ \sum_{n,m} e^{-i(n\sigma + m\sigma')} \{\alpha_n^\mu(\tau), \alpha_m^\nu(\tau)\} &= \frac{2\pi}{2\pi} \eta^{\mu\nu} \frac{d}{d\sigma} \sum_{n=-\infty}^{\infty} e^{in(\sigma - \sigma')} \\ \sum_{n,m} e^{-i(n\sigma + m\sigma')} \{\alpha_n^\mu(\tau), \alpha_m^\nu(\tau)\} &= \eta^{\mu\nu} \sum_{n=-\infty}^{\infty} in e^{in(\sigma - \sigma')} \end{aligned}$$

RHS:

$$\begin{aligned} &i\eta^{\mu\nu} \sum_{n=-\infty}^{\infty} n e^{in(\sigma - \sigma')} \\ &= i\eta^{\mu\nu} \sum_{n,m} (\delta_{n+m,0}) n e^{in\sigma + im\sigma'} \end{aligned}$$

Since $\delta(\sigma' - \sigma) = \delta(\sigma - \sigma')$ $e^i \rightarrow e^{-i}$

$$n \rightarrow -n, \quad m \rightarrow -m$$

$$\{\alpha_n^{\mu}, \alpha_m^{\nu}\} = i\eta^{\mu\nu}(-n)\delta_{n+m,0}$$

Similarly for $\bar{\alpha}^{\mu}$:

*We obtain

$$\begin{aligned}\{\alpha_n^{\mu}, \alpha_m^{\nu}\} &= \{\bar{\alpha}_n^{\mu}, \alpha_m^{\nu}\} = -in\eta^{\mu\nu}\delta_{n+m,0} \\ \{\alpha_n^{\mu}, \bar{\alpha}_m^{\nu}\} &= 0\end{aligned}$$

Now using $x_0^{\mu}, \alpha_0^{\mu}, \alpha_n^{\mu}(\tau), \bar{\alpha}_n^{\mu}(\tau), n \neq 0$
to describe the phase space (p^{μ}, x^{μ})

• Now, we obtain the E.O.M for $\alpha_n^{\mu}, \dots$

Recall $\ddot{X}^{\mu} = \dot{X}'^{\mu}$

$$\frac{\partial}{\partial \tau} \mathcal{P}^{\mu} = \sum_n \dot{\alpha}_n^{\mu} e^{-in\sigma}$$

$$\partial_{\sigma}(\mathcal{P}^{\mu}) = \frac{1}{\sqrt{2\alpha'}} (\ddot{X}^{\mu} + \dot{X}'^{\mu})$$

$$\partial_{\sigma}(\mathcal{P}^{\mu}) = \frac{1}{\sqrt{2\alpha'}} (\ddot{X}'^{\mu} + \dot{X}''^{\mu})$$

But $\ddot{X}^{\mu} = \dot{X}'^{\mu}$ so we find that

$$\partial_{\tau} \mathcal{P}^{\mu} = \partial_{\sigma} \mathcal{P}^{\mu}$$

$$\sum_n \dot{\alpha}_n^{\mu} e^{-in\sigma} = \sum_n \alpha_n^{\mu} (-in) e^{-in\sigma}$$

$$\dot{\alpha}_n^{\mu}(\tau) = -in \alpha_n^{\mu}(\tau)$$

$$\alpha_n^{\mu}(\tau) = \alpha_n^{\mu}(0) e^{-in\tau}$$

$$\bar{\alpha}_n^{\mu}(\tau) = \bar{\alpha}_n^{\mu}(0) e^{-in\tau}$$

Recall that $\frac{\dot{X}_0^{\mu}}{\alpha'} = \sqrt{\frac{2}{\alpha'}} \alpha_0^{\mu}(\tau)$

where $\alpha_0^{\mu}(\tau) =: \alpha_0^{\mu}(0)$

$$\Rightarrow \dot{X}_0^{\mu}(\tau) = \sqrt{2\alpha'} \alpha_0^{\mu}$$

$$X_0^{\mu}(\tau) = \sqrt{2\alpha'} \alpha_0^{\mu} \tau + q^{\mu}$$

Here $q^{\mu} = x^{\mu}(0)$

* Since

$$\dot{X}^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu + \dot{X}^{\mu})$$

$$\dot{\bar{X}}^\mu = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^\mu - \dot{X}^{\mu})$$

$$\Rightarrow \sqrt{2\alpha'} (\dot{X}^\mu - \dot{\bar{X}}^\mu) = 2\dot{X}^{\mu}$$

$$(\dot{X}^{\mu})^2 = \frac{\sqrt{2\alpha'}}{2} (\dot{X}^\mu - \dot{\bar{X}}^\mu)^2$$

$$= \frac{\sqrt{2\alpha'}}{2} \left[\sum_{n=-\infty}^{\infty} (\alpha_n^\mu \alpha_n^\mu e^{-in\tau} e^{-in\sigma} - \bar{\alpha}_n^\mu \bar{\alpha}_n^\mu e^{-in\tau} e^{in\sigma}) \right]$$

$$\frac{d}{d\sigma} \dot{X}^\mu = \text{RHS}$$

• If we now integrate this over the closed string:

$$\int_{-\pi}^{\pi} d\sigma \frac{d}{d\sigma} \dot{X}^\mu(\tau, \sigma) = \dot{X}^\mu(\tau, \sigma) \Big|_{\sigma=-\pi}^{\sigma=\pi}$$

$$+ \frac{\sqrt{2\alpha'}}{2} \sum_{n \neq 0} \left[\alpha_n^\mu \alpha_n^\mu e^{-in\tau} (e^{-in\pi} / -in) - \bar{\alpha}_n^\mu \bar{\alpha}_n^\mu e^{-in\tau} (e^{in\pi} / in) \right]$$

Recall: $\dot{X}^\mu(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \dot{X}_n^\mu(\tau) e^{-in\sigma}$

$$\Rightarrow \dot{X}^\mu(\tau, 0) = \sum_{n=-\infty}^{\infty} \dot{X}_n^\mu(\tau)$$

$\Rightarrow$ NO, instead we isolate $n=0$

$$\frac{\sqrt{2\alpha'}}{2} (\alpha_0^\mu - \bar{\alpha}_0^\mu) = 0 \quad \text{since } \alpha_0^\mu = \bar{\alpha}_0^\mu$$

$$\dot{X}(\tau, \sigma) = \dot{X}_0^\mu(\tau) + i \frac{\sqrt{2\alpha'}}{2} \sum_{n \neq 0} \frac{2}{n} \left[\alpha_n^\mu \alpha_n^\mu e^{-in(\sigma+\tau)} + \bar{\alpha}_n^\mu \bar{\alpha}_n^\mu e^{-in(\tau-\sigma)} \right]$$

Since $\dot{X}(\tau, \sigma) = \sum_{n=-\infty}^{\infty} \dot{X}_n^\mu(\tau) e^{-in\sigma}$ $n=0$

$$\dot{X}_0^\mu(\tau) = \dot{q}^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau$$

$$\text{Since } \frac{\sqrt{2\alpha'}}{2} p^\mu = \alpha_0^\mu \Rightarrow \alpha^\mu p^\mu = \sqrt{2\alpha'} \alpha_0^\mu$$

This can be written as

$$\begin{aligned} X(\tau, \sigma) &= X^L(\tau + \sigma) + X^R(\tau - \sigma) \\ X^L(\tau + \sigma) &= \frac{q}{2} + \alpha' \frac{p}{2} (\tau + \sigma) + i \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{\alpha_n}{n} e^{-in(\tau + \sigma)} \\ &\quad \text{"} \leftarrow - \sigma \text{"} \end{aligned}$$

We have only imposed the constraint

$$\partial_\mu (\sqrt{-g} g^{\alpha\beta} \partial_\beta x^\alpha) = 0$$

which was from the x^μ E.O.M. The $g_{\alpha\beta}$ E.O.M. yields

$$T_{\alpha\beta} = 0$$

$$T_{00} = T_{11} = \frac{1}{2} (\dot{X}^\mu \dot{X}^\nu + X'^\mu X'^\nu) \eta_{\mu\nu}$$

$$T_{01} = T_{10} = \dot{X}^\mu X'^\nu \eta_{\mu\nu}$$

Since

$$p^\mu = \frac{1}{2\alpha'} (\dot{X}^\mu + X'^\mu)$$

$$\begin{aligned} \therefore p^\mu p_\mu &= \frac{1}{2\alpha'} \eta_{\mu\nu} (\dot{X}^\mu + X'^\mu) (\dot{X}^\nu + X'^\nu) \\ &= \frac{1}{2\alpha'} \eta_{\mu\nu} (\dot{X}^\mu \dot{X}^\nu + 2 \dot{X}^\mu X'^\nu + X'^\mu X'^\nu) \end{aligned}$$

$$\alpha' p^\mu p_\mu = (T_{00} + T_{01})$$

or

$$\begin{aligned} p^\mu p_\mu &= \frac{1}{\alpha'} (T_{00} + T_{01}) \equiv \frac{1}{\alpha'} T_{++} \\ \dot{X}^\mu \dot{X}_\mu &= \frac{1}{\alpha'} (T_{00} - T_{01}) \equiv \frac{1}{\alpha'} T_{--} \end{aligned}$$

• Since $T_{\alpha\beta}$, as we shall see, breaks up into holomorphic & Anti-holomorphic terms $T_{zz} \equiv T(z)$ & $T_{\bar{z}\bar{z}} \equiv \bar{T}(\bar{z})$ with a mode expansion

$$T(z) = \sum_{n=-\infty}^{\infty} L_n z^{-n-2}$$

h_T : (Conformal weight) $h_T = 2$ ((2,0) tensor)

Requiring $T=0$ would implicitly imply that $L_n = \bar{L}_n = 0$

• In particular,

$$L_n = \oint \frac{dz}{2\pi i} z^{n+1} T(z)$$

$$= \oint \frac{dz}{2\pi i} z^{n+1} T(\bar{z})$$

So the relationship between $T(z)$ and $T_{++}(\sigma)$ is given by

$$z = e^{i(\tau + \sigma)}$$

$$T(z) \sim \frac{1}{\alpha'} T_{++}$$

$$\Rightarrow L_n = \frac{1}{2\pi\alpha'} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} T_{++} = \frac{1}{4\pi} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} p^2$$

$$= \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_m^\mu \alpha_{n-m}^\nu \eta_{\mu\nu}$$

and likewise for $\bar{L}_n$.

$$\begin{cases} \{L_n, L_m\} = -i(n-m)L_{n+m} \\ \{\bar{L}_n, \bar{L}_m\} = -i(n-m)\bar{L}_{n+m} \end{cases} \quad \left| \quad \begin{cases} \{L_n, \bar{L}_m\} = 0 \\ \forall n, m \end{cases} \right.$$

The Open String: $-\pi \leq \sigma \leq \pi$

* Must Account for Boundary Conditions.

$$X^\mu(\tau, \sigma) = \begin{cases} X^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ X^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

• Note we have a reflection about $\sigma=0$.

Consider $\sigma \rightarrow \sigma \in (-\infty, \infty)$

s.t. $X(\tau, \sigma) = X(\tau, \sigma + 2\pi)$. This is achieved via the fact

$$X^m(\tau, \pi) = X^m(\tau, -\pi)$$

$$\Rightarrow \frac{d}{d\sigma}(X^m(\tau, \pi)) = 0$$

So $X^m(\tau, \sigma)$ is continuous & passes a first derivative at $\sigma = \pm\pi$.

We can capture the constraints on T_{00}, T_{11}, T_{01} via the Equations:

$$(\mathcal{P}^\mu \mathcal{P}_\mu)(\tau, \sigma) = 0 \quad -\pi < \sigma < \pi$$

Requiring that each vanish separately.

$$\mathcal{P}^\mu(\tau, \sigma) = \frac{1}{2\alpha'} (\dot{X}^\mu(\tau, \sigma) + X'^\mu(\tau, \sigma)), \quad 0 \leq \sigma \leq \pi$$

* To find the extension of $\mathcal{P}^\mu(\tau, \sigma)$ we simply apply:

$$\mathcal{P}^\mu(\tau, \sigma) = \begin{cases} \mathcal{P}^\mu(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ \mathcal{P}^\mu(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

Q: Is this right? Let's find out:

$$\frac{d}{d\tau} X^m(\tau, \sigma) = \begin{cases} \dot{X}^m(\tau, \sigma) & 0 \leq \sigma \leq \pi \\ \dot{X}^m(\tau, -\sigma) & -\pi \leq \sigma \leq 0 \end{cases}$$

$\dot{X}^m$ is an EVEN f^m under $\sigma \rightarrow -\sigma$

$$\text{* However, } \frac{d}{d\sigma} X^m(\tau, \sigma) = \begin{cases} X'^m(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ -X'^m(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

under $\sigma \rightarrow -\sigma$:

$$X'^m(\tau, -\sigma) = \begin{cases} -X'^m(\tau, \sigma), & 0 \leq \sigma \leq \pi \\ X'^m(\tau, -\sigma), & -\pi \leq \sigma \leq 0 \end{cases}$$

So X'^m is ODD

Boundary Conditions:

$$\frac{d}{d\sigma} X^\mu(\tau, \sigma) = 0, \quad \sigma = 0, \sigma = \pi$$

Recall we had,

$$X^\mu(\tau, \sigma) = T^\mu(\tau) \cdot A(\sigma)$$

$$\Rightarrow \frac{d}{d\sigma} X^\mu = T^\mu(\tau) A'(\sigma) = 0 \quad (\sigma = 0, \pi)$$

It is therefore obvious to have

$$A'(\sigma) = \sin(n\sigma)$$

* The full expansion is therefore

$$X^\mu(\tau, \sigma) = \sum_{n=0}^{\infty} X_n^\mu(\tau) \cos(n\sigma) = X_0^\mu(\tau) + \sum$$

note

$$\begin{aligned} \sum_{n=0}^{\infty} X_n^\mu(\tau) \cos(n\sigma) &= \sum_{n=0}^{\infty} X_n^\mu \frac{1}{2} (e^{in\sigma} + e^{-in\sigma}) \\ &= \frac{1}{2} \sum_{n=0}^{\infty} X_n^\mu(\tau) e^{in\sigma} + \frac{1}{2} \sum_{n=0}^{\infty} X_n^\mu(\tau) e^{-in\sigma} \\ &= () + \frac{1}{2} (n \rightarrow -n) \end{aligned}$$

$$= () + \frac{1}{2} \sum_{n=-\infty}^{-1} X_{-n}^\mu(\tau) e^{in\sigma}$$

* Note: X^μ is even under $\sigma \rightarrow -\sigma$

as such $X_{-n}^\mu(\tau) = X_n^\mu(\tau)$

$\Rightarrow \therefore$

$$X^\mu(\tau, \sigma) = X_0^\mu(\tau) + \sum_{n=1}^{\infty} X_n^\mu(\tau) e^{in\sigma}$$

* We use orthogonal f_n^s to solve for the coefficients of $\sum: X_n^\mu(\tau)$

$$\int_{-\pi}^{\pi} d\sigma X^\mu(\tau, \sigma) \cos(m\sigma) = \sum_{n=-\infty}^{\infty} X_n^\mu(\tau) \int_{-\pi}^{\pi} \cos(n\sigma) \cos(m\sigma) d\sigma$$

$$\begin{aligned} n=0: \int_{-\pi}^{\pi} d\sigma X^\mu(\tau, \sigma) \cos(m\sigma) \\ = \sum_{n=-\infty}^{\infty} X_n^\mu(\tau) \int_{-\pi}^{\pi} \cos(n\sigma) \cos(m\sigma) d\sigma \end{aligned}$$

$$\int_{-\pi}^{\pi} d\sigma \cos(n\sigma) \cos(m\sigma) = \begin{cases} 0 & n \neq m \\ \pi & n = m \neq 0 \\ 2\pi & n = m = 0 \end{cases}$$

* $n=0$, then $m=0$

$$X_0^M(\tau) / 2\pi = \int_{-\pi}^{\pi} d\sigma \underline{X}^M(\tau, \sigma)$$

$$\text{or } X_0^M(\tau) = \left(\frac{2}{\pi}\right) \int_{-\pi}^{\pi} d\sigma \underline{X}^M(\tau, \sigma)$$

$n=m \neq 0$:

$$X_n^M(\tau) = \frac{2}{\pi} \int_{-\pi}^{\pi} d\sigma' \cos(n\sigma') \underline{X}^M(\tau, \sigma'), \quad n \neq 0$$

* The same Equation of motion holds for the open string, $\partial^2 \underline{X}^M = 0$

$$(-\partial_\tau^2 + \partial_\sigma^2) \underline{X}^M(\tau, \sigma)$$

$$(\partial_\tau^2 + \partial_\sigma^2) \left\{ \sum_{n=-\infty}^{\infty} X_n^M(\tau) \cos(n\sigma) \right\}$$

$$= \sum_{n=-\infty}^{\infty} \left\{ (-\ddot{X}_n^M(\tau) - n^2 X_n^M(\tau)) \cos(n\sigma) \right\}$$

We could argue that at each "level" the above $\{ \}$ must vanish but a better way to see this is to integrate using orthogonal fn's. [TRIVIAL]:

$$\therefore \ddot{X}_n^M(\tau) + n^2 X_n^M(\tau) = 0$$

* $n \geq 1$ we have an infinite # of harmonic oscillators plus a Center of mass coordinate: $X_0^M(\tau) = q^M + p^M \tau$

$\rightarrow$ Still need to introduce the constraint.

$$\tau_{03} = 0$$

$$p^M(\tau, \sigma) \equiv \sum_{n=-\infty}^{\infty} \alpha_n^M(\tau) e^{in\sigma}$$

$$\alpha_n^\mu(\tau) = \frac{2}{2\pi} \int_{-\pi}^{\pi} d\sigma e^{in\sigma} \mathcal{P}^\mu(\tau, \sigma)$$

* Using $\tilde{X}(-\sigma) = \tilde{X}(\sigma)$ and $\tilde{X}'(-\sigma) = -\tilde{X}'(\sigma)$

$$\mathcal{P}^\mu(\tau, \sigma) - \mathcal{P}^\mu(\tau, -\sigma) = \frac{2\tilde{X}'^\mu(\tau, \sigma)}{\sqrt{2\alpha'}}.$$

$$\int_{-\pi}^{\pi} \frac{d\sigma}{2\pi} \frac{2\tilde{X}'^\mu(\tau, \sigma)}{\sqrt{2\alpha'}} = \int_{-\pi}^{\pi} d\sigma \{ \mathcal{P}^\mu(\tau, \sigma) - \mathcal{P}^\mu(\tau, -\sigma) \}$$

$$\begin{aligned} X^\mu(\tau, \sigma) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} \{ \alpha_n^\mu(\tau) e^{-in\sigma} - \alpha_{-n}^\mu(\tau) e^{in\sigma} \} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma \sum_{n=-\infty}^{\infty} \{ \alpha_n^\mu(\tau) e^{-in\sigma} - \alpha_{-n}^\mu(\tau) e^{in\sigma} \} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} d\sigma \left(\alpha_0^\mu(\tau) - \alpha_0^\mu(\tau) \right) \\ &\quad + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{2}{n} (\alpha_n^\mu(\tau) - \alpha_{-n}^\mu(\tau)) e^{-in\sigma} \end{aligned}$$

$$\therefore \underline{\tilde{X}}^\mu(\tau, \sigma) = X_0^\mu(\tau) + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{2}{n} (\alpha_n^\mu(\tau) - \alpha_{-n}^\mu(\tau)) e^{-in\sigma}$$

E.O.M: $(-\partial_\tau^2 + \partial_\sigma^2) \tilde{X}^\mu(\tau, \sigma) = 0$

$\Rightarrow X_0^\mu(\tau)$ satisfies this separately so that

$$\dot{\alpha}_n^\mu(\tau) = -in\alpha_n^\mu(\tau) \Rightarrow \alpha_n^\mu(\tau) = e^{-in\tau} \alpha_n^\mu(0)$$

Note, at the level of operators this expression is given by

$$\mathcal{P}^\mu = \mathcal{P}^{\mu}$$

or

* The zero mode term:

$$\begin{aligned} \frac{1}{2\tilde{X}^\mu/\sqrt{2\alpha'}} &= (\mathcal{P}^\mu(\tau, \sigma) + \mathcal{P}^\mu(\tau, -\sigma)) \\ \frac{2\tilde{X}_0^\mu/\sqrt{2\alpha'}}{2\pi} &= \frac{2}{2\pi} \int_{-\pi}^{\pi} d\sigma (\mathcal{P}^\mu(\tau, \sigma) + \mathcal{P}^\mu(\tau, -\sigma)) \\ &= 2\alpha_0^\mu \end{aligned}$$

$$\tilde{X}_0^\mu(\tau) = \sqrt{2\alpha'} \alpha_0^\mu$$

$$q^\mu \equiv X^\mu(0)$$

$$X_0^\mu(\tau) = q^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau$$

$$\dot{X}^\mu(\tau, \sigma) = \dot{q}^\mu + \sqrt{2\alpha'} \alpha_0^\mu + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} (\alpha_n^\mu e^{-in(\tau+\sigma)} + \alpha_{-n}^\mu e^{-in(\tau-\sigma)})$$

Open-string Expansion

Using the relation $P^\mu(\sigma) = \sqrt{2\alpha'} \pi \dot{X}^\mu + \frac{\dot{X}^\mu \dot{X}^\mu}{\sqrt{2\alpha'}}$
we find

$$\{P^\mu(\sigma), P^\nu(\sigma')\} = 2\pi\alpha' \frac{d}{d\sigma} \delta(\sigma - \sigma')$$

* Again, expanding in Fourier comp's

$$\{\alpha_n^\mu, \alpha_m^\nu\} = -in \delta_{n+m,0} \eta^{\mu\nu}$$

• Total momentum: $P^\mu \equiv \int_0^\pi d\sigma P^\mu = \frac{\dot{X}_0^\mu}{2\alpha'} = \frac{\alpha_0^\mu}{\sqrt{2\alpha'}}$

Note: $\{\dot{X}^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$

$$\int_0^\pi d\sigma \{\dot{X}^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu}$$

$$\int_0^\pi d\sigma \dot{X}^\mu(\sigma) = \dot{X}_0^\mu \quad \text{and} \quad \text{so}$$

$$\{q^\mu, p^\nu\} = \eta^{\mu\nu}$$

E.M. Tensor & Angular Mom. of the string:

* Poincaré Symmetry:

$$-\pi^{\mu\nu} = -\frac{\partial \mathcal{L}}{\partial(2\alpha' X_{\mu\nu})} \quad (\text{translational current})$$

$$(M^{\mu\nu})_{\mu\nu} = X^\mu \frac{\partial \mathcal{L}}{\partial(2\alpha' X^\nu)} - (\text{ang. mom.}) = X^\mu \pi^{\nu\alpha} - X^\nu \pi^{\mu\alpha}$$

Using the EoL-lag. eq's these currents are conserved:

$$\partial^\alpha \pi_\alpha^\mu = 0, \quad \partial^\alpha M_\alpha^{\mu\nu} = 0$$

$$S = \frac{1}{2\pi\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

$g_{\alpha\beta} \approx \eta_{\alpha\beta}$, then

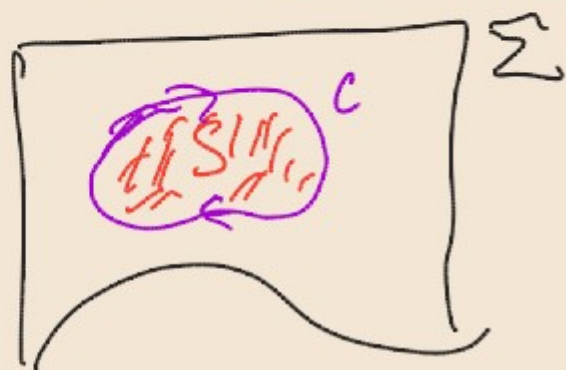
$\Leftrightarrow$

$$\pi_\alpha^\mu = \frac{1}{2\pi\alpha'} \partial_\alpha X^\mu$$

likewise

$$M_\alpha^{\mu\nu} = \frac{1}{2\pi\alpha'} (X^\mu \partial_\alpha X^\nu - X^\nu \partial_\alpha X^\mu)$$

* Take a closed curve C on the worldsheet with surface S within



Then, using Stokes Thm

$$\int_C (\pi_0^\mu d\sigma + \pi_1^\mu d\tau) = - \int_S d\tau d\sigma \partial^\alpha \pi_\alpha^\mu = 0$$

In terms of differential forms:

$$\hat{\pi}^\mu = G_{\alpha\beta} \pi^{\alpha\beta} d\zeta^\beta$$

Flow of energy/momentum through any curve C on the worldsheet

$\Rightarrow$ Flow of E & B through the ends of the string are zero.

A Classical solution of the open string:

$\rightarrow$ Find explicit sol^{ns} for

$$\partial_\alpha \partial^\alpha x^\mu = 0 \quad \& \quad T_{\alpha\beta} = 0$$

$$\bullet \quad t \equiv x^0 = \tau d, \quad d \in \mathbb{R}$$

$$T_{00} \Leftrightarrow \frac{1}{2} (\dot{x}^\mu \dot{x}^\nu + x^{\mu'} x^{\nu'}) \eta_{\mu\nu} = 0$$
$$= \dot{x}^0 \dot{x}^0 + \sum_{i=1}^{D-1} \dot{x}^i \dot{x}^i - x^{0'} x^{0'} + \dots$$

